

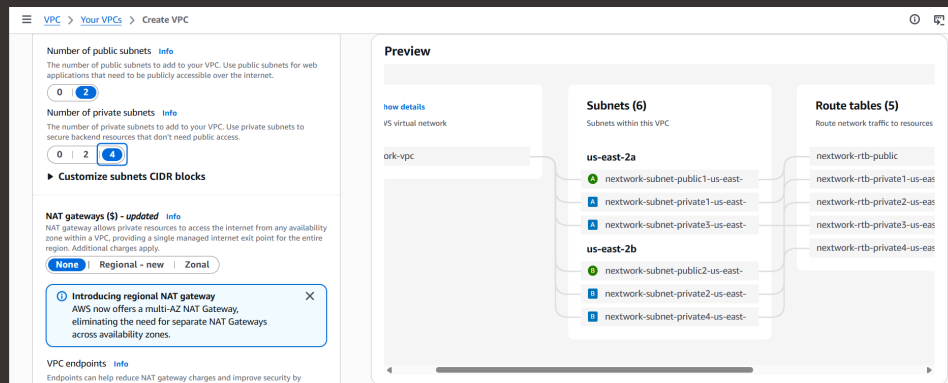


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Launching VPC Resources



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources. It is useful because it gives you control over networking and security, allowing you to decide how resources connect to each other and to the internet.

How I used Amazon VPC in this project

I used Amazon VPC to create a secure cloud network, set up public and private subnets, configure route tables and an internet gateway, and apply security groups and network ACLs to control traffic to my resources.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how many different networking pieces like subnets, route tables, security groups, and network ACLs have to work together just to allow or block traffic.



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This project took me...

This project took me thirty minutes to complete.



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Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to it straight from your computer using a remote access method such as SSH.

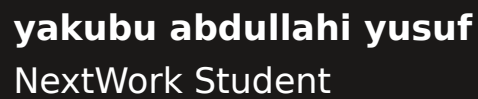
SSH is a key method for directly accessing a VM

SSH traffic means network communication used to securely connect to and manage a remote computer over the internet using SSH.

To enable direct access, I set up key pairs

Key pairs are a set of security credentials made up of a public key and a private key that are used to securely log in to EC2 instances.

A private key's file format defines how the key is stored, and my key was saved as a .pem file.



Launching a public server

I had to change my EC2 instance's networking settings by editing its VPC, security group and subnet configuration, and that the instance was placed in the public subnet.

Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
VPC ID vpc-016d9b867bab64aff (NextWork VPC) 🔗			Subnet ID subnet-0413f5192d99a28ec (NextWork Public Subnet) 🔗		Availability zone us-east-2a	
Availability zone ID use2-az1			Outpost ID -			
▼ IP addresses Info						
Public IPv4 address 18.188.41.124 open address 🔗			Private IPv4 addresses 10.0.0.187		IPv6 addresses -	
Secondary private IPv4 addresses -			Carrier IP addresses (ephemeral) -			
▼ Hostname and DNS Info						
Public DNS -			Private IP DNS name (IPv4 only) ip-10-0-0-187.us-east-2.compute.internal		IPv4-only IP based name: A record only -	
Dualstack - IP based name: A and AAAA record -			IPv6-only - IP based name: AAAA record only -		Public hostname type -	
Private hostname type IP name: ip-10-0-0-187.us-east-2.compute.internal			Use RBN as guest OS hostname Disabled		Answer RBN DNS hostname IPv4 Disabled	

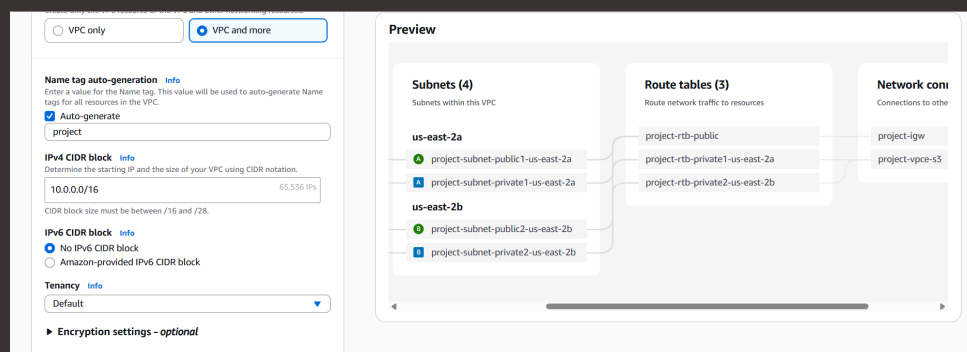


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used the Amazon VPC wizard to automatically create the VPC and its networking components instead of configuring everything manually.

A VPC resource map is diagram that helps you see your VPC, subnets, route tables, internet gateways, and security controls in one view.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because VPCs are isolated from each other by default, even if their IP address ranges overlap.



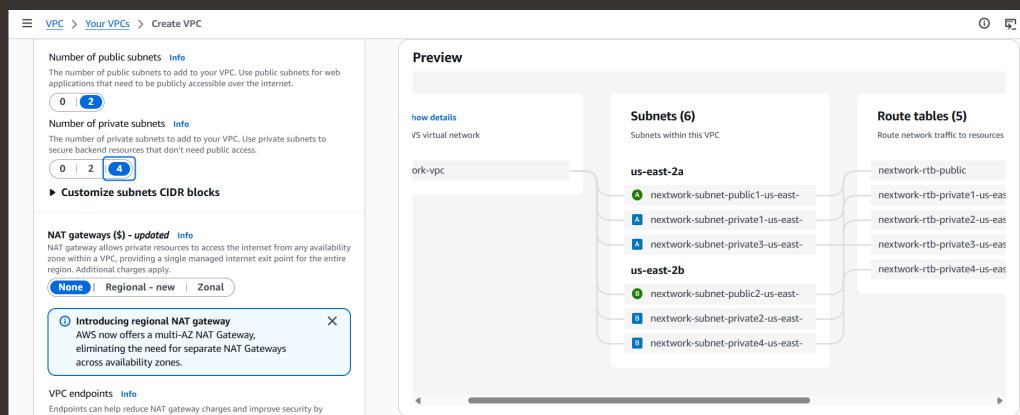


Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options, one or two. This was because I could only choose one or two public subnets because the wizard creates one public subnet per Availability Zone.

The set up page also offered to create NAT gateways, NAT gateways let private resources send traffic out, but block incoming internet traffic.





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